

Expressive Communication in Families

**Familiarity and Accuracy in an
Interactive Emotion Recognition
Task**

Kye Palmer, Mark Kohler, Nicole Nelson

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I respectfully acknowledge the Turrbal people, who are the traditional owners of the lands we meet on today, and the Kaurna People, whose lands this research was conducted on, and where I live and work.



We rely on emotional expressions every day to interpret how people are feeling...



...but how good are we at reading the expressions of those closest to us?

Why emotion recognition matters

- Emotion understanding develops through early social interaction
- Parents help shape this development
 - Through emotion socialisation, parents model, label, and respond to emotions, shaping how children learn to express and interpret their feelings (Eisenberg et al., 1998; Morris et al., 2007; Castro et al., 2015)
- Children may be better at expressing and interpreting emotions with familiar people, though this is largely unexplored
- We need more interactive, ecologically valid methods

**How well can parents and children
interpret each other's emotional
expressions during an interactive task –
and does familiarity play an important
role?**

Study 1 – parent-child dyads

- Recruited 45 parent-child dyads (children aged 4 to 12)
- Recruited and tested at the South Australian Museum (N = 32) and in a research lab (N = 13)
- Participants completed an object guessing game with each other, then with video clips of unfamiliar parents and children.
- Expressions were video recorded, and were later shown to naïve adults

What we predicted

1. Parents would perform better in the expression recognition task with their own child, compared to other unknown children
 - Parents average task performance would improve with child age
2. Children would perform better in the expression recognition task with their own parents, compared to other unknown parents
3. *Based on prior work using this paradigm*, familiar parents will perform better at recognising expressions of their own child compared to naïve parents, and both will perform better than non-parents

Method



8 boxes



Stickers (**Happy**)



Spiders (**Fear**)



Broken Photo (**Sad**)



Poop (**Disgust**)

2 of each object

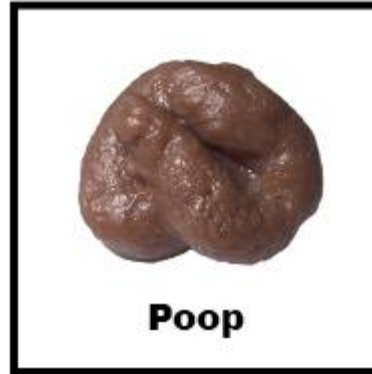
Method



Stickers



Spider



Poop



Broken Photo

Familiar Person Box Task

1



Round 1

Parent is the expresser
Child is the guesser



Round 2

Child is the expresser
Parent is the guesser

Unfamiliar Person Box Task

2



Parent watches videos of
other age-matched children
expressing

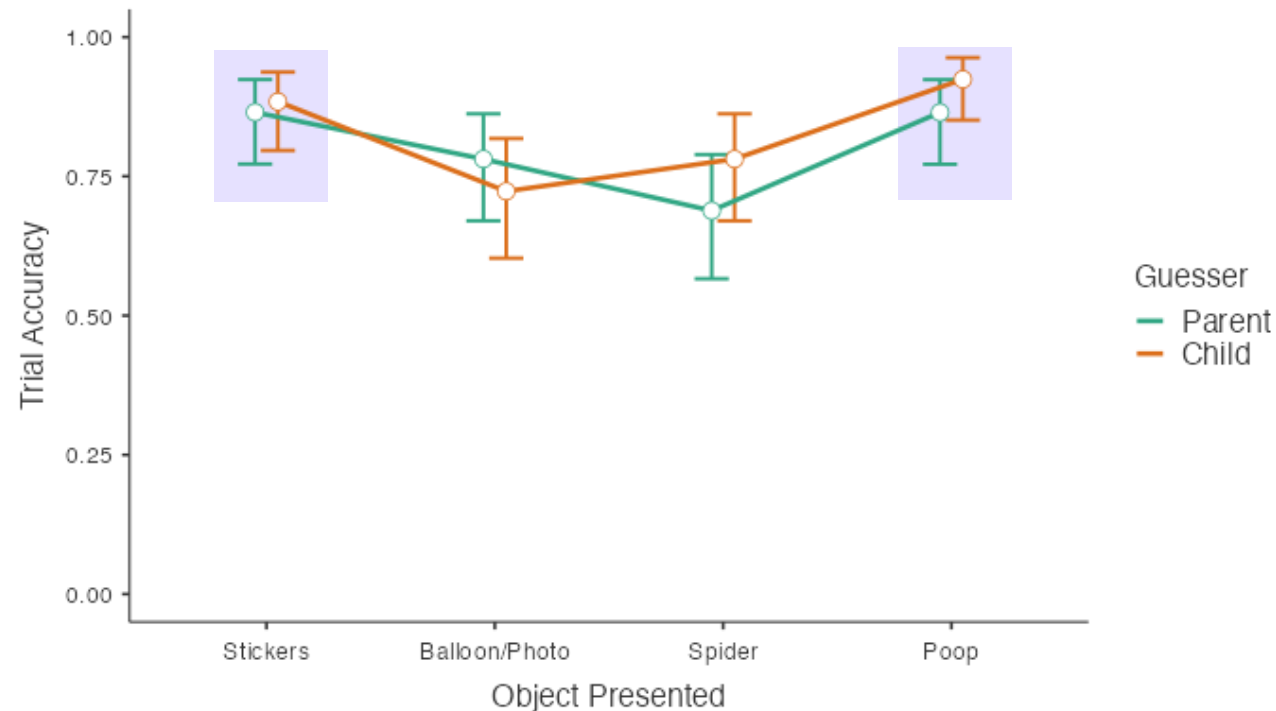


Child watches videos of
other parents expressing
(to age-matched kids)

Results – familiar trial

Parents and children perform similarly with each other

Parent and child trial accuracy across object presented



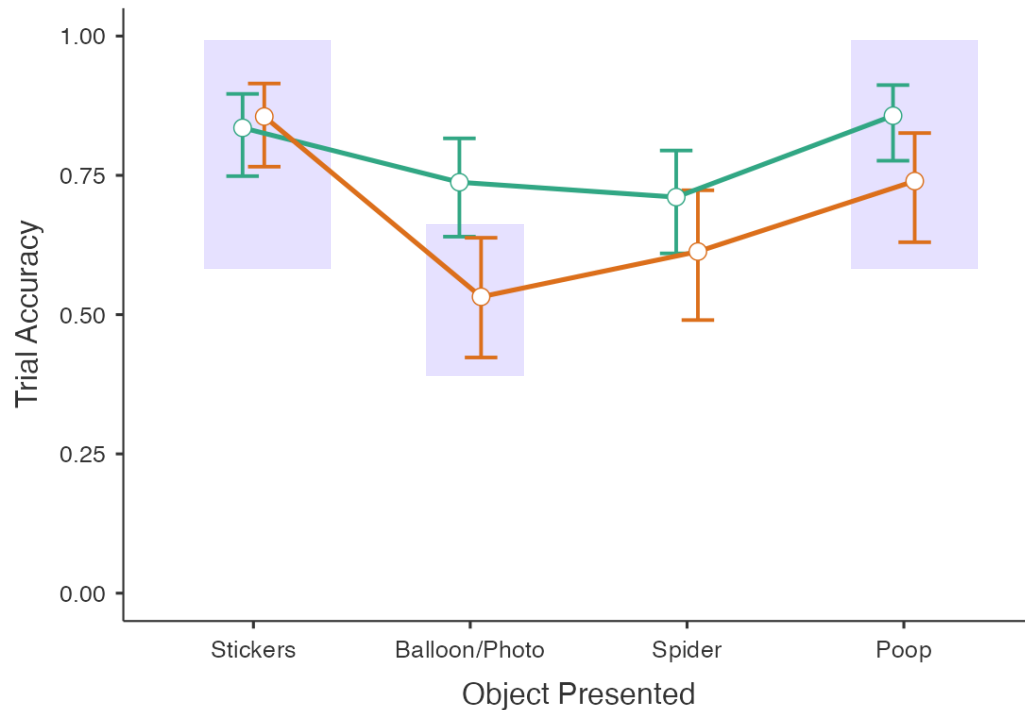
- No significant difference between **parents and children** in their **accuracy** with each other
- **Object presented** influenced recognition
- **Child age** was a weak but significant predictor of performance ($p = 0.049$)

Results – unfamiliar and familiar trial

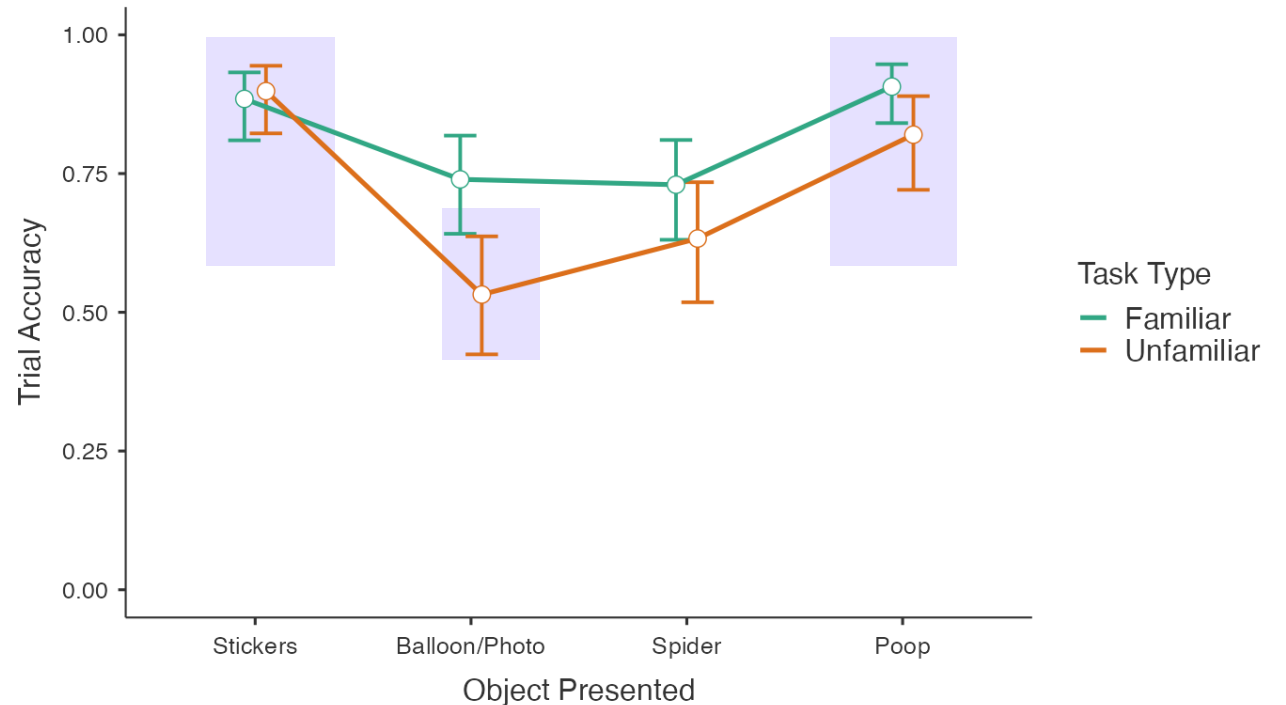
Familiarity improves accuracy

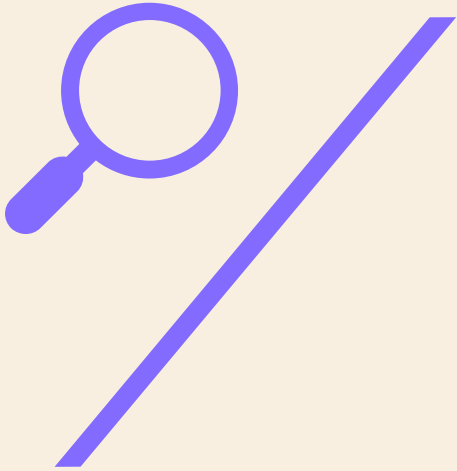
- **Object type** influenced recognition

Parents performance in familiar and unfamiliar tasks, across object



Children's performance in familiar and unfamiliar tasks, across object





Naïve Adult Raters

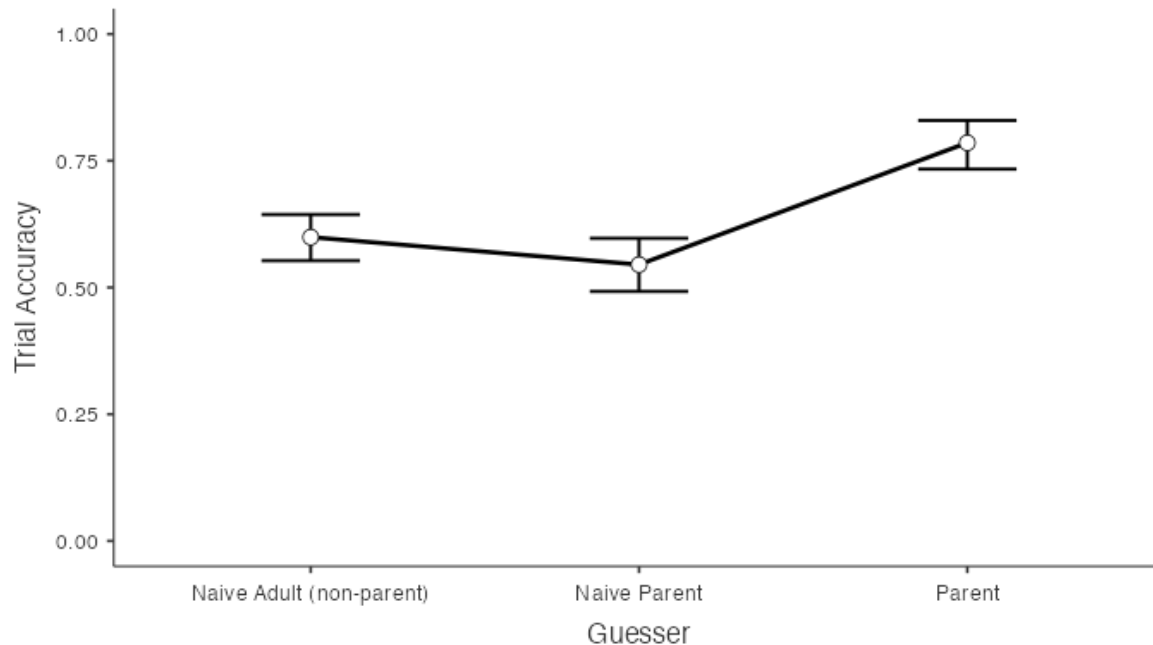
To explore the familiarity effect further, video recordings from Study 1 were presented to naive adults

We recruited 69 parents (of similarly aged children) and 68 non-parents

Naïve Adult Raters

Parent-specific advantage

Mean trial accuracy across non-parents, naïve parents, and task-familiar parents

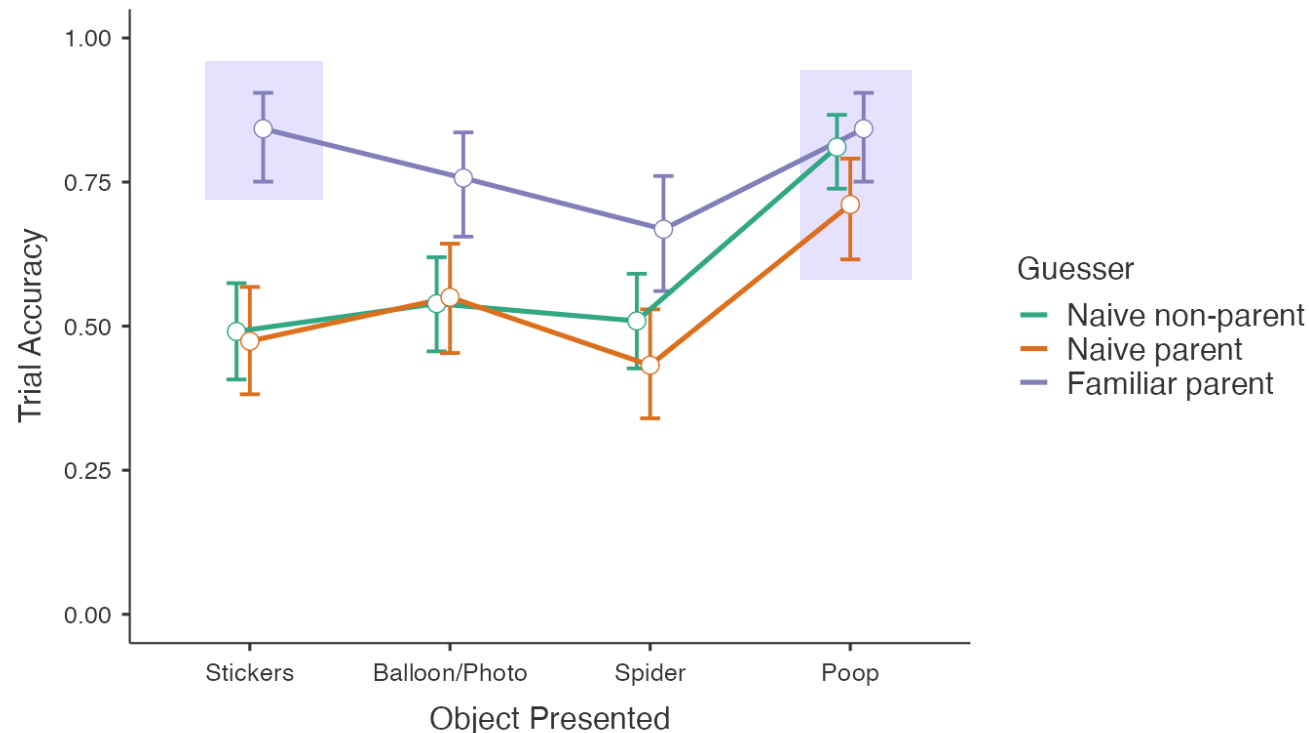


- **Familiar Parents** were significantly more accurate than both **naïve parents** *and* **non-parents**
- No significant difference between **naïve parents** and **non-parents**

Results

Parent-specific advantage
...for happy expressions!

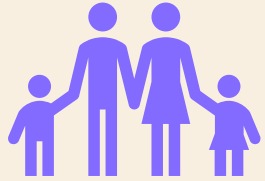
Mean trial accuracy for naïve adults and task-familiar parents, across object presented



- **Familiar parents** were significantly more accurate than both **naïve parents** and **non-parents** on sticker trials
- Previous research has highlighted that children struggle to purposefully express happy in a way that is recognisable to adults (Fong et al., 2020)
- Everyone did quite well with disgust!

Summary & Conclusions

- Familiarity enhances expression recognition
 - Parents and children performed better with each other
- Parents display an advantage in recognising their own child's expressions
 - Familiar parents outperformed ALL naïve adults (including other parents)
- No strong age effects
 - Accuracy did not consistently improve with age. Relationship may be more important than the developmental stage?



Study 2 - Siblings

Following on from Study 1, we extended the box task paradigm to siblings.

Why siblings?

- With age, parents often report that they begin to spend less time with their children (Åman-Back & Björkqvist, 2004; Wikle & Cullen, 2023)
- Children with siblings will often spend more time with each other than with other children their own age (Campione-Barr, 2017; Dunifon et al., 2017)
- Siblings may be actively stimulating each other's development of emotion understanding and expression in similar ways to other social behaviours (Kramer, 2014)



Study 2 – Siblings

- Recruited 44 parent-sibling triads (children aged 4 to 13)
- Recruited and tested at the South Australian Museum (N = 41) and in a research lab (N = 3)

Guessers sit opposite the expresser,
on either side of a small divider



After each expresser opens 8
boxes, the positions rotate

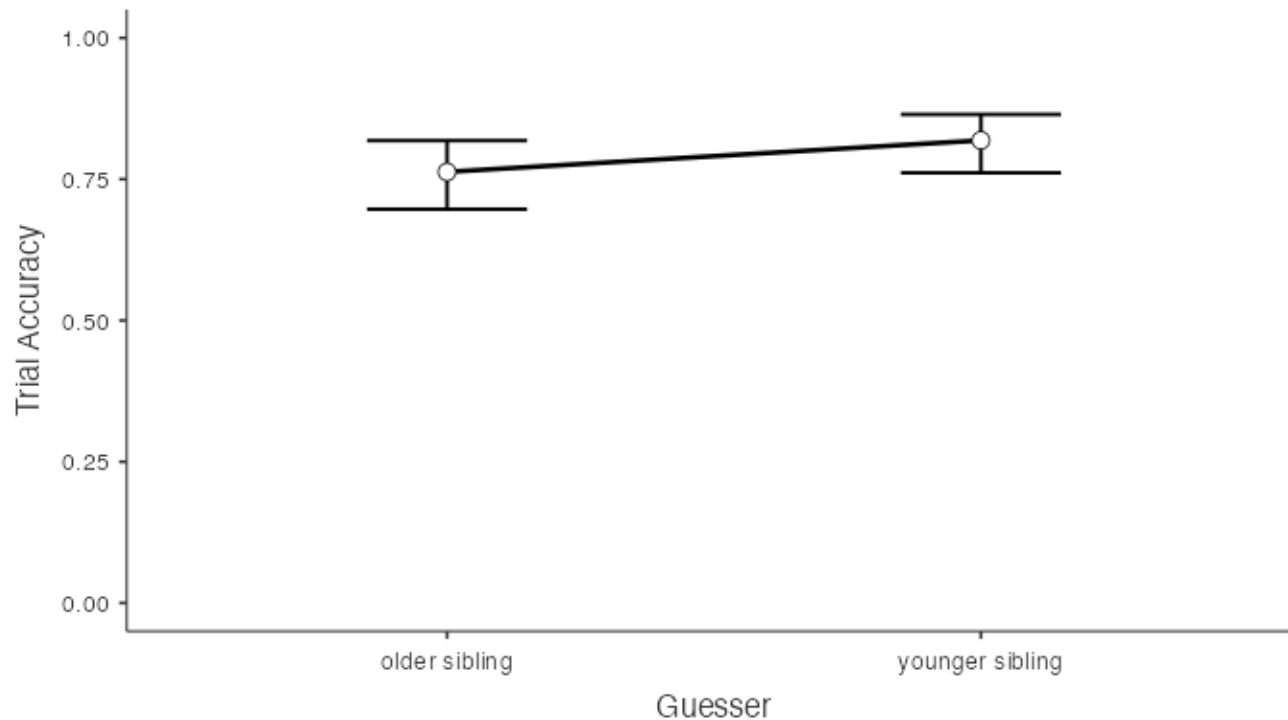
Siblings then watch videos of other unfamiliar children completing the same task



Results – Sibling Familiar Trial

Siblings are similarly accurate with each other, regardless of birth order

Familiar trial accuracy between younger and older siblings

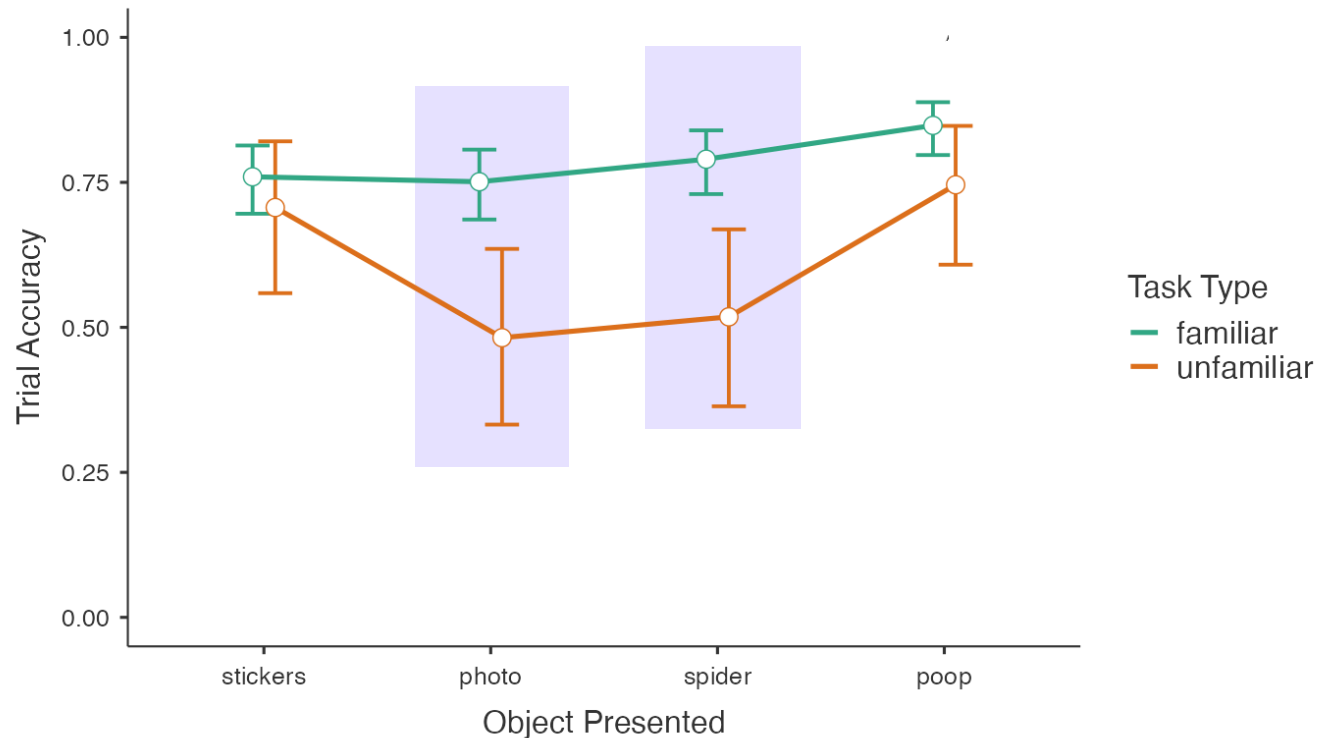


→ Older sibling ‘expertise’ did not translate into being better at reading their sibling’s expressions.

Results – Sibling Unfamiliar Trial

Siblings are more accurate with each other, than with unfamiliar children

Sibling accuracy by object and task type



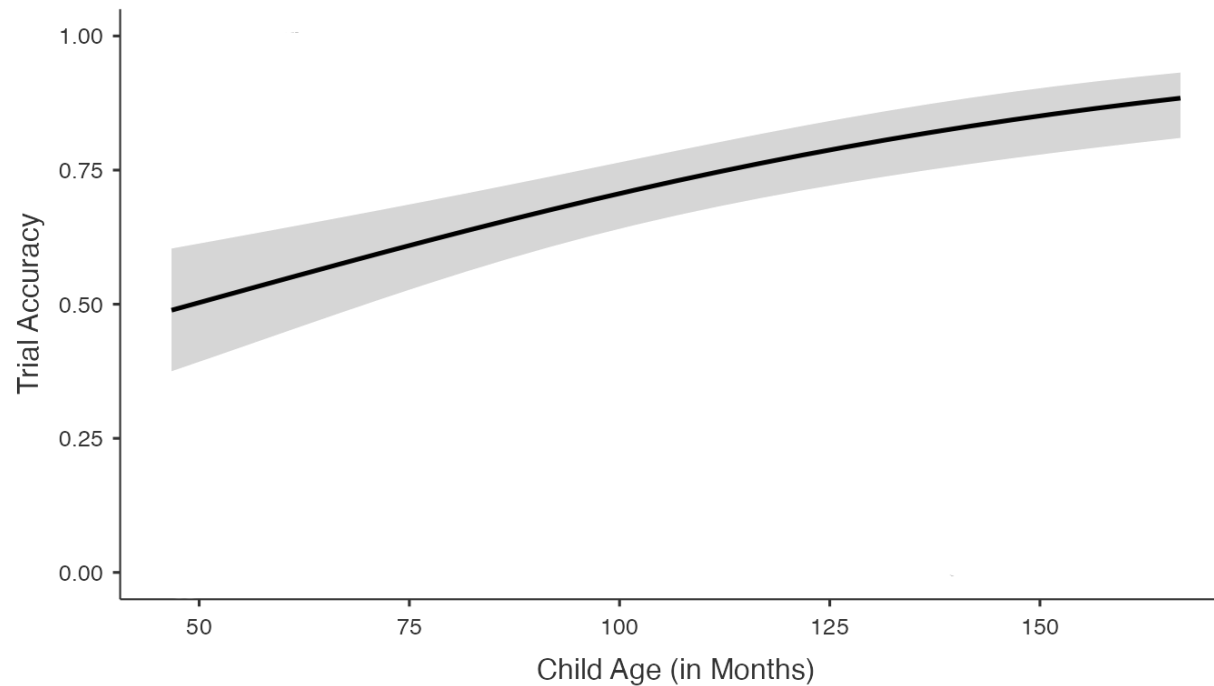
→ Siblings performed better in the task with each other, than when viewing unfamiliar children.

→ By object, siblings were more accurate with each other for broken photo and spider, but no significant difference was found for stickers or poop.

Results – Age

Older children make more accurate guesses,
regardless of familiarity

Sibling changes in accuracy across age



→ The age effect held across
both familiar and unfamiliar
trials.

Summary & Conclusions

- Familiarity effect extends to sibling relationships
 - Siblings perform better with each other, than with unfamiliar children
- Performance improves with age
 - Children get better at recognising emotional expressions from their sibling and other children as they get older.
- Siblings have been an underexplored context for emotional development.
 - Kids spend more time with siblings than same age-peers
 - This relationship may play a distinct role in how emotional understanding develops
 - It's a good dynamic to explore further to better understand emotional understanding

Future Directions

- Currently underway in our lab, we are looking more closely at children's expressions and people's interpretations of these.
 - Exploring the similarity of expressions among siblings
 - Using eye-tracking technology to investigate what areas of the face and body people attend to when interpreting children versus adults' expressions in the box task.

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Connect with me!



(Bonus points if you connect
with me **and** chat post-docs
with me 😊)

Building on previous findings using dynamic stimuli

Fong et al. (2020)

Children guessed experimenter's expressions in an interactive game

Key Findings

- Recognition accuracy improved with age



Buttle et al. (2020)

Children and parents complete the same game, guessing each other's expressions. Naïve adults guessed children's expressions.

Key Findings

- Parents were more accurate than naïve adults. Potential effect of familiarity.
- No effect of child age.



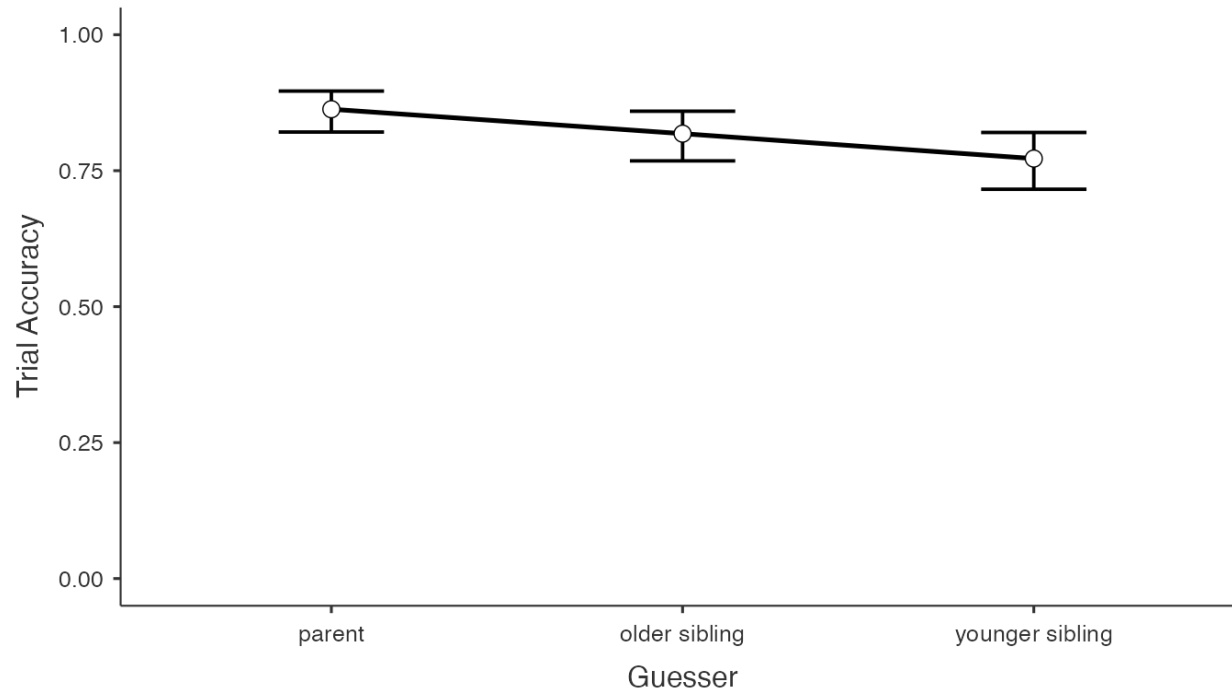
The Current Studies

Further investigating this familiarity effect with parents and siblings

Results – siblings and parents together

Parents outperform younger siblings, but not older siblings

Trial accuracy across parents, older siblings, and younger siblings



→ Parents were more accurate than younger siblings at recognising expressions.

→ Older siblings did not differ significantly from parents or younger siblings.

Validation of Naïve Adult Comparison

- In our main analyses, familiar parents completed the expression recognition task as a live interaction with their own child, seeing each object twice across eight trials.
- Naïve adult raters viewed randomised video clips from multiple children, meaning they could encounter the same object more than once and were less able to use a process of elimination strategy.
- To examine whether this difference contributed to the familiarity advantage observed in Study 2, we recruited a new sample of naïve adults who each viewed eight videos from a single child, with two examples of each object, matching the structure of the familiar parent task.

Validation of Naïve Adult Comparison

With our Naïve adults who viewed a single child as our reference group, familiar parents were significantly more accurate overall. This advantage did not vary across object types.

